

π 的一个级数表示

Horst Alzer

经典的数 π 不仅在几何学中,也在其他的数学领域,例如分析学、数论、统计学中有重要的作用,并且它在工程学和一些物理学的领域有着有趣的应用. 关于它的相关性,许多作者已经彻底研究了 π 的性质. 特别地,我们可以找到 π 及其相关量的许多不同寻常的级数、乘积和积分表示. 关于这个主题的详细信息,请参考 Arndt 和 Haenel 的专著 [1], 研究论文 Sofo [4] 以及在那里所引的文献.

本文的目的是提出 π 的一个级数表示, 它把 π 与 Leibniz (莱布尼茨) 级数的下述部分和联系起来:

$$T_k = \sum_{j=0}^k \frac{(-1)^j}{2j+1} \quad (k = 0, 1, 2, \dots).$$

优美的公式

$$\frac{\pi}{4} = \lim_{k \rightarrow \infty} T_k = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \quad (1)$$

是众所周知的结果, 于 1682 年首次由 G. W. Leibniz 发表. Borwein 等 [2] 对 $\frac{\pi}{4} - T_k$ 提供了一个包含 Euler (欧拉) 数的渐近公式, Rattaggi [3] 对 $\frac{\pi}{4} - T_k$ 提供了多种上界和下界.

这里是我们的 π 的表示.

定理 我们有

$$\pi = 32 \sum_{k=0}^{\infty} (-1)^{k+1} \frac{4k^2 + 8k + 1}{(2k-1)(2k+1)(2k+3)(2k+5)} T_k^2. \quad (2)$$

证明 首先, 我们证明

$$a_k = b_k \quad (k = 0, 1, 2, \dots), \quad (3)$$

其中

$$a_k = \frac{1}{(2k+3)^2} + \frac{1}{(2k+5)^2} - \frac{1}{2k+3} + \frac{1}{2k+5}$$

和

$$b_k = \frac{2}{2k+5} T_k^2 + \frac{4}{(2k+3)(2k+5)} T_{k+1}^2 - \frac{2}{2k+3} T_{k+2}^2.$$

令 $k \geq 0$. 我们有

$$\begin{aligned} a_k &= \frac{2}{(2k+3)(2k+5)} \left(\frac{1}{2k+3} - \frac{1}{2k+5} \right) \\ &= \frac{2(-1)^k}{(2k+3)(2k+5)} \left(\frac{(-1)^k}{2k+3} + 2T_{k+1} - 2T_{k+1} + \frac{(-1)^{k+1}}{2k+5} \right) \end{aligned}$$

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作者的邮箱地址是 h.alzer@gmx.de.

$$\begin{aligned}
&= \frac{2}{2k+5} \frac{(-1)^k}{2k+3} \left(\frac{(-1)^k}{2k+3} + 2T_{k+1} \right) + \frac{2}{2k+3} \frac{(-1)^{k+1}}{2k+5} \left(2T_{k+1} + \frac{(-1)^k}{2k+5} \right) \\
&= \frac{2}{2k+5} (T_k - T_{k+1})(T_k + T_{k+1}) + \frac{2}{2k+3} (T_{k+1} - T_{k+2})(T_{k+1} + T_{k+2}) \\
&= \frac{2}{2k+5} (T_k^2 - T_{k+1}^2) + \frac{2}{2k+3} (T_{k+1}^2 - T_{k+2}^2) \\
&= b_k.
\end{aligned}$$

令

$$A_n = \sum_{k=0}^n (-1)^k a_k \quad \text{和} \quad B_n = \sum_{k=0}^n (-1)^k b_k. \quad (4)$$

则

$$\begin{aligned}
A_n &= \sum_{k=0}^n (-1)^k \left(\frac{1}{(2k+3)^2} + \frac{1}{(2k+5)^2} \right) + \sum_{k=0}^n (-1)^k \left(\frac{1}{2k+5} - \frac{1}{2k+3} \right) \\
&= \frac{1}{9} + \frac{(-1)^n}{(2n+5)^2} + \frac{1}{3} + \frac{(-1)^n}{2n+5} - 2 \sum_{k=0}^n \frac{(-1)^k}{2k+3}
\end{aligned} \quad (5)$$

和

$$\begin{aligned}
B_n &= 2 \sum_{k=0}^n \frac{(-1)^k}{2k+5} T_k^2 + 4 \sum_{k=0}^n \frac{(-1)^k}{(2k+3)(2k+5)} T_{k+1}^2 - 2 \sum_{k=0}^n \frac{(-1)^k}{2k+3} T_{k+2}^2 \\
&= 16 \sum_{k=1}^n (-1)^{k+1} \frac{4k^2 + 8k + 1}{(2k-1)(2k+1)(2k+3)(2k+5)} T_k^2 \\
&\quad + 2(-1)^n \frac{4n^2 + 20n + 17}{(2n+1)(2n+3)(2n+5)} T_{n+1}^2 + 2(-1)^{n+1} \frac{1}{2n+3} T_{n+2}^2 - \frac{22}{45}.
\end{aligned} \quad (6)$$

由于

$$\sum_{k=0}^n \frac{(-1)^k}{2k+3} = 1 - T_{n+1},$$

则从 (1) 和 (5) 我们得到

$$\lim_{n \rightarrow \infty} A_n = \frac{1}{9} + \frac{1}{3} - 2 \left(1 - \frac{\pi}{4} \right) = \frac{\pi}{2} - \frac{14}{9}. \quad (7)$$

从 (1) 和 (6) 我们得到

$$\lim_{n \rightarrow \infty} B_n = 16 \sum_{k=1}^{\infty} (-1)^{k+1} \frac{4k^2 + 8k + 1}{(2k-1)(2k+1)(2k+3)(2k+5)} T_k^2 - \frac{22}{45}. \quad (8)$$

利用 (3), (4), (7) 和 (8), 给出

$$\frac{\pi}{2} - \frac{14}{9} = -\frac{22}{45} + 16 \sum_{k=1}^{\infty} (-1)^{k+1} \frac{4k^2 + 8k + 1}{(2k-1)(2k+1)(2k+3)(2k+5)} T_k^2.$$

这样, 即得

$$\pi = 2 \left(\frac{14}{9} - \frac{22}{45} - \frac{16}{15} \right) + 32 \sum_{k=0}^{\infty} (-1)^{k+1} \frac{4k^2 + 8k + 1}{(2k-1)(2k+1)(2k+3)(2k+5)} T_k^2.$$

这就导致 (2). ■

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图片来源 图 1–6 及题图由作者提供.

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(陆柱家 译 童欣 校)